

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: Hydrofluosilicic Acid 23-25% (AFCO 8896)

Product Code: AFCO 8896

Intended Use of the Product

Use of the Substance/Mixture: Industrial chemical. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

Alex C. Fergusson, LLC.
800 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Met. Corr. 1	H290
Acute Tox. 4 (Oral)	H302
Acute Tox. 3 (Dermal)	H311
Skin Corr. 1	H314
Eye Dam. 1	H318

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



Signal Word (GHS-US)

: Danger.

Hazard Statements (GHS-US)

: H290 - May be corrosive to metals
H302 - Harmful if swallowed
H311 - Toxic in contact with skin
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage

Precautionary Statements (GHS-US)

: P234 - Keep only in original container.
P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P270 - Do not eat, drink or smoke when using this product.
P280 - Wear protective glove, protective clothing, eye protection, face protection.
P301+P312+P330 - IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell. Rinse mouth.
P301+P330+P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303+P361+P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340+P310 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a Poison Control Center/doctor.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a Poison Control Center/doctor.

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P362 - Take off contaminated clothing and wash before reuse..
P390 - Absorb spillage to prevent material damage.
P405 - Store locked up.
P406 - Store in corrosive resistant container with a resistant inner liner.
P501 - Dispose of contents/container to local, regional, national, territorial, provincial, and international regulations.

Other Hazards

Other Hazards: None known.

Unknown Acute Toxicity (GHS-US): Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Mixture

Name	Product identifier	% (w/w)	Classification (GHS-US)
Fluorosilicic acid	(CAS No) 16961-83-4	30-50	Met. Corr. 1, H290 Acute Tox 4 (Oral), H302 Skin Corr. 1A, H314
Hydrofluoric acid	(CAS No) 7664-39-3	0.1-1	Acute Tox. 2 (Oral), H300 Acute Tox. 1 (Dermal), H310 Acute Tox. 2 (Inhalation: vapour), H330 Skin Corr. 1A, H314

Actual concentration is withheld as a trade secret

Any Concentration shown as a range is due to batch variation.

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Move out of dangerous area. Consult a physician. Show this safety data sheet to the doctor in attendance. Symptoms of poisoning may appear several hours later. Do not leave the victim unattended.

Inhalation: If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a physician.

Skin Contact: Immediate medical treatment is necessary as untreated wounds from corrosion of the skin heal slowly and with difficulty. Take victim immediately to hospital. If on skin, rinse well with water. If on clothes, remove clothes.

Eye Contact: Small amounts splashed into eyes can cause irreversible tissue damage and blindness. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Continue rinsing eyes during transport to hospital. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.

Ingestion: Keep respiratory tract clear. Do NOT induce vomiting. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician. Take victim immediately to hospital.

Most Important Symptoms and Effects Both Acute and Delayed

Not available.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If exposed or concerned, get medical advice and attention.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water mist, carbon dioxide, foam, dry powder.

Unsuitable Extinguishing Media: High volume water jet.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: No dangerous reaction known under conditions of normal use.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Use water spray or fog for cooling exposed containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: No hazardous combustion products are known.

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Other information: Do not allow run-off from fire fighting to enter drains or water courses. Collect contaminated fire extinguishing water separately. Do not discharge fire extinguishing water into drains.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid breathing (vapors, mist, dust). Avoid all contact with skin, eyes, or clothing.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Prevent entry to sewers and public waters. Avoid release to the environment. Prevent further leakage or spillage if safe to do so.

If the product contaminates rivers, lakes, or drains inform appropriate authorities.

Methods and Material for Containment and Cleaning Up

For Containment and Cleaning Up: Neutralize with chalk, alkali solution or ammonia. Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: Smoking, eating and drinking should be prohibited in the application area. To avoid spills during handling keep bottle on a metal tray. Dispose of rinse water in accordance with local and national regulations.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling. Do not breathe vapors/mists. Avoid contact with skin and eyes. For personal protection see Section 8.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in a dry, cool and well-ventilated place. Keep container tightly closed when not in use. Keep/Store away from extremely high or low temperatures & incompatible materials. Prevent unauthorized access.

Incompatible Materials: Do not store near acids.

Special Rules on Packaging: Keep only in original container. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

Specific End Use(s): Industrial chemical. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Hydrofluoric acid (7664-39-3)		
Mexico	OEL Ceiling (mg/m ³)	2.5 mg/m ³
Mexico	OEL Ceiling (ppm)	3 ppm
USA ACGIH	ACGIH TWA (ppm)	0.5 ppm
USA ACGIH	ACGIH Ceiling (ppm)	2 ppm
USA ACGIH	ACIGH (TWA) (ppm)	0.5 ppm(fluorine)
USA ACGIH	ACIGH (Ceiling) (ppm)	2 ppm (fluorine)
USA OSHA Z-2	OSHA PEL (TWA) (ppm)	3 ppm
USA OSHA PO	OSHA(PEL) TWA (ppm)	3 ppm (fluorine)
USA OSHA PO	OSHA (STEL) (ppm)	6 ppm (fluorine)
USA NIOSH	NIOSH REL (TWA) (mg/m ³)	2.5 mg/m ³
USA NIOSH	NIOSH REL (TWA) (ppm)	3 ppm
USA NIOSH	NIOSH REL (ceiling) (mg/m ³)	5 mg/m ³

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USA NIOSH	NIOSH REL (ceiling) (ppm)	6 ppm
USA IDLH	US IDLH (ppm)	30 ppm
Ontario	OEL Ceiling (ppm)	2 ppm
Ontario	OEL TWA (ppm)	0.5 ppm
Québec	PLAFOND (mg/m ³)	2.6 mg/m ³
Québec	PLAFOND (ppm)	3 ppm

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas. General and local exhaust ventilation is recommended to maintain exposures levels below recommended limits.

Personal Protective Equipment: Protective goggles. Protective clothing. Gloves. Insufficient ventilation: wear respiratory protection. Face shield.



Materials for Protective Clothing: Chemically resistant materials and fabrics. Corrosion proof clothing.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles. Wear face shield for abnormal processing problems.

Skin and Body Protection: Wear suitable protective clothing.

Respiratory Protection: Where concentrations are above recommended limits or are unknown, appropriate respiratory protection should be worn. Follow OSHA respirator regulations (29 CR 1910.134) and use NIOSH/MSHA approved respirators. Protection provided by air purifying respirators against exposures to any hazardous chemical is limited. Use a positive pressure air supplied respirator if there is any potential for uncontrolled release, exposure levels are unknown, or any other circumstances where air purifying respirators may not provide adequate protection.

Thermal Hazard Protection: Wear suitable protective clothing.

Other Information: When using, do not eat, drink or smoke. Avoid contact with skin, eyes, & clothing.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid.
Appearance	: Clear, colorless.
Odor	: Pungent.
Odor Threshold	: Not data available.
pH	: 1-2
Relative Evaporation Rate (butylacetate=1)	: Not data available.
Melting Point	: -20°C (-4°F)
Freezing Point	: -20°C (-4°F)
Boiling Point	: 136 - 163°C (277 - 325°F)
Flash Point	: Does not flash.
Auto-ignition Temperature	: Not data available.
Decomposition Temperature	: Not data available.
Flammability (solid, gas)	: Not data available.
Lower Flammable Limit	: Not data available.
Upper Flammable Limit	: Not data available.
Vapor Pressure	: Not data available.
Relative Vapor Density at 20°C	: Not data available.
Relative Density	: 1.2 @ 25°C (77°F) Reference substance: (water = 1)
Specific Gravity	: 1.236
Solubility	: Miscible in water.
Partition coefficient: n-octanol/water	: Not data available.

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SECTION 10: STABILITY AND REACTIVITY

Reactivity: No dangerous reaction known under conditions of normal use.

Chemical Stability: Stable under normal conditions.

Possibility of Hazardous Reactions: No decomposition if stored and applied as directed.

Conditions to Avoid: Keep away from heat, flame, sparks and other ignition sources.

Incompatible Materials: Glass. Strong oxidizing agents.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects -

Product:

Acute Oral Toxicity: Acute toxicity estimate: 500.1 mg/kg.

Acute Inhalation Toxicity: Acute toxicity estimate: 50.01 mg/l, Exposure time: 4 h, Test atmosphere: vapor.

Acute Dermal Toxicity: Acute toxicity estimate: 500.05 mg/kg

Components:

Hydrofluoric acid (7664-39-3):

Acute Oral Toxicity: Assessment: The component/mixture is highly toxic after single ingestion.

Remarks: No data available.

Acute Inhalation Toxicity: LC50 (Rat): 1610 ppm, Assessment: The component/mixture is highly toxic after short term inhalation.

Acute Dermal Toxicity: Assessment: The component/mixture is extremely toxic after single contact with skin.

Remarks: No data available.

Skin Corrosion/Irritation

Product:

Remarks: Extremely corrosive and destructive to tissue.

Components:

Fluorosilicic acid (16961-83-4):

Species: Rat

Result: Causes burns.

Hydrofluoric acid (7664-39-3):

Species: Rabbit.

Result: Causes severe burns.

Serious Eye Damage/Irritation

Product:

Remarks: May cause irreversible eye damage.

Components:

Hydrofluoric acid (7664-39-3):

Species: Rabbit,

Result: Risk of serious damage to eyes.

Germ Cell Mutagenicity

Components:

Hydrofluoric acid (7664-39-3):

Genotoxicity in Vitro: Test Type: Ames test, Species: Salmonella typhimurium Result: negative.

Genotoxicity in Vivo: Test Type: In vivo micronucleus test, Species: Mouse Result: negative.

Germ Cell Mutagenicity - Assessment: Tests on bacterial or mammalian cell cultures did not show mutagenic effects.

Carcinogenicity

Components:

Hydrofluoric acid (7664-39-3):

Species: Rat

NOAEL: 25 mg/kg bw/day

Carcinogenicity - Assessment: Not classifiable as a human carcinogen.

IARC No component of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed

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human carcinogen by IARC.

OSHA No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.

NTP No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

Reproductive Toxicity

Components:

Hydrofluoric acid (7664-39-3):

Reproductive Toxicity – Assessment: Fertility classification not possible from current data.

Teratogenicity - Assessment: Embryotoxicity classification not possible from current data.

Further Information

Product:

Remarks: No data available.

SECTION 12: ECOLOGICAL INFORMATION

Toxicity Not available.

Components:

Hydrofluoric acid (7664-39-3)	
LC50 Fish 1	No data available.
EC50 Daphnia 1	97 mg/l (Exposure time: 48 h - Species: Daphnia magna [Water flea])
ECr50 (algae)	No data available.

Persistence and Degradability

Hydrofluoric acid (7664-39-3)	
Persistence and Degradability	No data available.

Bioaccumulative Potential

Hydrofluoric acid (7664-39-3)	
Bioaccumulative Potential	No data available.

Mobility in Soil No data available.

Product:

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Other Adverse Effects

Ozone-Depletion Potential: Regulation: 40 CFR Protection of Environment; Part 82 Protection of Stratospheric Ozone - CAA Section 602 Class I Substances, Remarks: This product neither contains, nor was manufactured with a Class I or Class II ODS as defined by the U.S. Clean Air Act Section 602 (40 CFR 82, Subpt. A, App.A + B).

Additional Ecological Information: No data available.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

Ecology - Waste Materials: This material is hazardous to the aquatic environment. Keep out of sewers and waterways.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT : UN1778, FLUOROSILICIC ACID, 8, II.

14.3 In Accordance with IATA : UN1778, FLUOROSILICIC ACID, 8, II.

14.2 In Accordance with IMDG : UN1778, FLUOROSILICIC ACID, 8, II.

SECTION 15: REGULATORY INFORMATION

EPCRA - Emergency Planning and Community Right-to-Know Act

CERCLA Reportable Quantity

Components	CAS-No.	Component RQ (lbs)	Calculated product RQ (lbs)
Hydrofluoric acid	7664-39-3	100	10001
Hydrochloric acid	7647-01-0	5000	*

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*: Calculated RQ exceeds reasonably attainable upper limit.

SARA 304 Extremely Hazardous Substances Reportable Quantity

Components	CAS-No.	Component RQ (lbs)	Calculated product RQ (lbs)
Hydrofluoric acid	7664-39-3	100	10001
Hydrochloric acid	7647-01-0	5000	*

*: Calculated RQ exceeds reasonably attainable upper limit.

SARA 311/312 Hazards

: Corrosive to metals.
Acute toxicity (any route of exposure)
Skin corrosion or irritation.
Serious eye damage or eye irritation.

SARA 302

: This material does not contain any components with a section 302 EHS TPQ.

SARA 313

: This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

Clean Air Act

This product does not contain any hazardous air pollutants (HAP), as defined by the U.S. Clean Air Act Section 112 (40 CFR 61).

This product does not contain any chemicals listed under the U.S. Clean Air Act Section 112(r) for Accidental Release Prevention (40 CFR 68.130, Subpart F).

This product does not contain any chemicals listed under the U.S. Clean Air Act Section 111 SOCM Intermediate or Final VOC's (40 CFR 60.489).

Clean Water Act

The following Hazardous Substances are listed under the U.S. Clean Water Act, Section 311, Table 116.4A:

7647-01-0	Hydrochloric acid
7664-39-3	Hydrofluoric acid

The following Hazardous Chemicals are listed under the U.S. Clean Water Act, Section 311, Table 117.3:

7647-01-0	Hydrochloric acid
7664-39-3	Hydrofluoric acid

This product does not contain any toxic pollutants listed under the U.S. Clean Water Act Section 307

Massachusetts Right To Know

16961-83-4	Fluorosilicic Acid
7647-01-0	Hydrochloric acid
7664-39-3	Hydrofluoric acid

Pennsylvania Right To Know

7732-18-5	Water
16961-83-4	Fluorosilicic Acid
7647-01-0	Hydrochloric acid
7664-39-3	Hydrofluoric acid

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California Prop 65

: This product does not contain any chemicals known to State of California to cause cancer, birth defects, or any other re-productive harm.

The components of this product are reported in the following inventories:

TSCA	: On TSCA Inventory.
DSL	: All components of this product are on the Canadian DSL.
AICS	: On the inventory, or in compliance with the inventory.
NZIoC	: Not in compliance with the inventory.
ENCS	: On the inventory, or in compliance with the inventory.
KECI	: On the inventory, or in compliance with the inventory.
PICCS	: On the inventory, or in compliance with the inventory.
IECSC	: On the inventory, or in compliance with the inventory.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date : 03/02/2021

Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

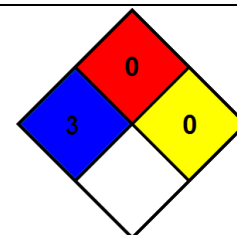
GHS Full Text Phrases:

Met. Corr. 1	Corrosive to metals Category 1.
Acute Tox. 2 (Oral)	Acute toxicity (oral) Category 2.
Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4.
Acute Tox. 1 (Dermal)	Acute toxicity (dermal) Category 1.
Acute Tox. 3 (Dermal)	Acute toxicity (dermal) Category 3.
Acute Tox. 2 (Inhalation: vapour)	Acute toxicity (inhalation: vapour) Category 2.
Skin Corr. 1A	Skin corrosion/irritation Category 1A.
Eye Dam 1	Serious eye damage/eye irritation Category 1.
H290	May be corrosive to metals.
H300	Fatal if swallowed.
H302	Harmful if swallowed.
H310	Fatal in contact with skin.
H311	Toxic in contact with skin.
H314	Causes severe skin burns and eye damage.
H318	Causes serious eye damage.
H330	Fatal if inhaled.

NFPA Health Hazard : 3 - Short exposure could cause serious temporary or residual injury even though prompt medical attention was given.

NFPA Fire Hazard : 0 - Materials that will not burn.

NFPA Reactivity : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

Health : 3 Serious Hazard - Major injury likely unless prompt action is taken and medical treatment is given.

Flammability : 0 - Minimal Hazard.

Physical Hazard : 4 - Extreme Hazard.

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Party Responsible for the Preparation of This Document

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This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)